Anticipation as Imagination: Emotion-Aware and Simulated Planning in Social Robotics

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Abstract. We present an architecture for social robots that integrates *foundation models* for emotion-aware action selection with anticipatory simulation in Unity. Emotional cues guide the generation of candidate actions, which are virtually tested to predict outcomes and avoid failures. This process enhances decision-making, proactive explainability, and user trust. The approach, validated in some project scenarios, improves behavioural robustness and can be extended to other adaptive and transparent robotic applications.

Keywords: Social robotics · Foundation models · Anticipation · Explainable AI

1 Introduction

The development of intelligent social robots requires the ability to dynamically adapt to the operating context and the characteristics of the user, especially when it comes to making decisions in complex and unpredictable environments. In this scenario, planning actions at runtime is a prerequisite to ensure flexible and personalised behaviour. However, the reactive generation of plans alone offers no guarantees for the effectiveness or consistency of the selected actions with respect to the goals of the interaction.

To address this limitation, we propose the introduction of an action anticipation mechanism, understood as an internal and preemptive simulation of the effects of a behaviour before its execution. This mechanism allows the robot to autonomously evaluate the plausibility and desirability of an action and provides internal feedback that is useful for decision making, selecting more effective strategies and avoiding undesirable outcomes. Anticipation can also provide an information base for the creation of explicit explanations that improve the transparency of the robot's behaviour and increase the user's trust in the system. This approach is part of a broader line of research that has identified perceptual anticipation as one of the key mechanisms for autonomy and awareness in artificial systems. Previous work (see [3, 2, 4]) has proposed the concept of perception loop, a cyclical structure in which the robot makes predictions about the outcome of its actions and compares expectations with what it perceives, thus initiating an iterative process of adaptation and learning. The perception loop represents a first attempt to formalise the link between perception, action and expectation within a cognitive architecture for robots that are aware of their actions.

The work presented in this paper is in continuity with this theoretical framework and proposes an integrated architecture that utilises recent foundation models [1] to support and extend the robot's anticipation capability. In particular, the architecture combines a module to identify actions based on the user's emotional state with a module to simulate candidate actions in a virtual environment and provides an internal evaluation mechanism to guide the selection of the most appropriate behaviour. The solution was developed using some scenarios of the projects currently under development by the authors and tested in simulated environments. The aim was to explore the benefits of anticipation both for conscious planning and for the explainability of robot interaction.

2 Anticipation for Explainability and Trust in Runtime Planning

Anticipation is here defined as the system's ability to internally simulate and assess the consequences of an action before execution. This work extends the perception loop model proposed in previous research, moving from a reactive post-execution comparison between expectations and perceptions to a deliberative, proactive evaluation. In our approach, simulation becomes an integral component of decision-making: it functions as an internal oracle, predicting outcomes under given conditions and guiding the selection of the most suitable behaviour according to the user's goals and emotional state. This process supports proactive explainability, enabling the robot to justify chosen actions, account for discarded alternatives, and ground decisions in a structured knowledge base that links states, actions, contexts, and effects. The proposed architecture (Fig. 1) integrates emotional analysis, candidate action generation, and anticipatory validation through simulation in a single operational flow and was mainly tested in scenarios where a robot engages in artistic performances alongside human actors.

Emotion-Driven Action Identification. The first part of the architecture is focused on interpreting the user's emotions and translating those emotional states into a set of candidate actions. The main input comes from the analysis of a music clip captured in real time, processed by the Music2Emotion framework to identify sound patterns and convert them into one or more prevailing emotions. Once the emotion has been identified, the system integrates this information with additional data related to the scene in progress, such as the role assigned to the robot, the description of the situation and the expected duration of the animation. User feedback on the previous scene is also considered, in order to maintain continuity and narrative consistency.

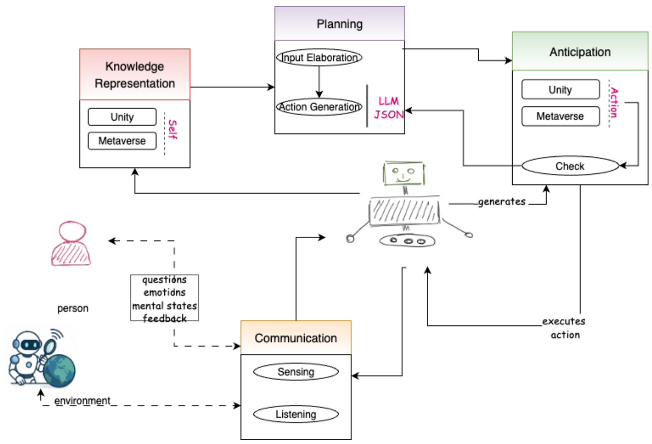

Fig. 1. Architecture and process for anticipating, representing and evaluating actions.

The determination of the most appropriate actions takes place through a Large Language Model (LLM) orchestrated using LangChain [6][5] and supported by symbolic reasoning modules. The model jointly evaluates the detected emotion, the user profile and the scenic context, generating an ordered sequence of movements and behaviors. Each action is described in a structured JSON format, in which each robot's joint is mapped to a specific position and a precise time interval, so as to allow subsequent interpolation to obtain continuous and dynamic movements. The emotional analysis is carried out through Music2Emotion, supported by the Transformer MERT model, trained on heterogeneous music datasets and capable of providing scores of emotional valence and intensity.

Motor sequences empathically consistent with the emotional state detected are generated and each decision is accompanied with an explicit motivation. For example, in the presence of emotions of sadness, the system may propose large, slowed movements to convey a sense of calm. Finally, the modular structure of the architecture and the adoption of interoperable formats such as JSON facilitate scalability and integration with heterogeneous robotic platforms.

Simulated Anticipation in Unity. The second part of the architecture takes as input the candidate actions, already encoded in JSON format, and verifies their feasibility through a simulated anticipation process. The goal is to detect potential critical issues in advance, such as collisions between robot parts or unwanted interactions with the surrounding environment, preventing such problems from occurring during real execution. The simulation is conducted in Unity, where a Digital Twin of the robot is used. During virtual execution, the system collects information on the outcomes of actions, recording any errors or failures, and uses this data to guide an automatic replanning process.

The replanning takes place through the use of an LLM which, starting from the feedback data and specially designed prompts, generates proposals for changes to the initial actions. The simulation and correction cycle is repeated until the actions are free of criticality then their execution on the real robot is authorized. The integration between Unity and ROS2, mediated by Python modules dedicated to orchestration and logical control, allows for smooth communication between the different components of the system. This approach ensures high operational safety, minimizing the risk of mechanical failure or unwanted behavior. In addition to increase the robustness of robotic behavior, anticipatory simulation allows the system to provide transparent explanations for its choices, such as: "Action A was not executed because it was unsuccessful in the simulation; action B was chosen as an alternative". In this way, prior validation not only improves the reliability of the system, but also contributes to its ability to generate well-founded explanations, fostering user understanding and trust.

3 Conclusion and Future Work

The proposed architecture integrates emotion-driven action selection and anticipatory simulation within a modular framework, enabling robots to predict, evaluate, and justify their actions before execution. This improves adaptability, explainability, and reliability, while fostering user trust.

Future work will raise the abstraction level of actions, integrate real sensors and robots, and extend anticipation to complex multi-agent social contexts.

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